## Part I. Biological cleavage and products

Percentages are study-specific efficiencies, not universal enzyme rate constants. Where a source reports only preference, spacing, or structure dependence, that is stated instead of converted into a percentage.

| RNase | Biological cleavage | mzLib cleavage | 3' biological product | 3' formula | 5' biological product | 5' formula | Occurrence / frequency evidence | Sources |
|---|---|---|---|---|---|---|---|---|
| RNase T1 | Cleaves after G in ssRNA (GpN). | G\| (full) | 3'-P via cyclic intermediate | H2O4P | 5'-OH | O-3P-1 | G sites observed throughout defined test RNA under optimal digestion; no missed cuts reported. | E006,E015 |
| RNase T2 | ssRNA endonuclease; little base specificity. | G\|,C\|,A\|,U\| (full) | 3'-P via cyclic intermediate | H2O4P | 5'-OH | O-3P-1 | Single-strand specific; no strong base preference; no universal rate reported. | E007,E025 |
| RNase A | Cleaves after unpaired C or U. | C\|,U\| (full) | 3'-P via cyclic intermediate | H2O4P | 5'-OH | O-3P-1 | C and U sites; C reported about 2x faster than U in one summary; structure and conditions also matter. | E008,E028 |
| RNase 1 | Mammalian pancreatic-type RNase; ssRNA cleavage 3' to pyrimidines. | G\|,C\|,A\|,U\| (full) | 3'-P via cyclic intermediate | H2O4P | 5'-OH | O-3P-1 | No universal site rate reported; mammalian RNase 1 is a distinct enzyme from E. coli RNase I. | E030,E031 |
| RNase I | E. coli endoribonuclease; largely sequence-independent on ssRNA. | G\|,C\|,A\|,U\| (full) | 3'-P via cyclic intermediate | H2O4P | 5'-OH | O-3P-1 | Broad ssRNA cleavage; no universal site rate reported. Ca2+ can support dsRNA activity, outside this ssRNA scope. | E009 |
| RNase PhyM (7 M urea) | Cleaves Up-N and Ap-N under 7 M urea, 50 C, pH 5. | A\|,U\| (full) | 2',3'-cyclic-P observed | O3P | Not reported | | Almost exclusive and uniform at Up-N/Ap-N under stated sequencing conditions; no percentage reported. | E010,E029 |
| RNase PhyM (no urea) | Cleaves Up-N, Ap-N, and Gp-N without urea. | A\|,U\|,G\| (full) | 2',3'-cyclic-P observed | O3P | Not reported | | Gp-N is additionally attacked without urea; relative efficiencies not reported. | E010,E029 |
| RNase 4 | Cleaves after U in U-A and U-G ssRNA sites. | U\|A,U\|G (full) | Mixed: linear 3'-P and 2',3'-cyclic-P | H2O4P,O3P | 5'-OH | O-3P-1 | UA is slightly preferred over UG; no absolute site percentage reported. | E011,E018 |
| RNase U2 | Purine-selective cleavage after A or G; A preferred. | G\|,A\| (full) | 3'-P via cyclic intermediate | H2O4P | 5'-OH | O-3P-1 | A > G >> C > U reported; no universal rate. | E012,E013 |
| Cusativin | Cleaves primarily at Cp[A/U/G] and UpA; additional UpU/ApU activity is reported. | C\| (full) | Mixed: linear 3'-P and 2',3'-cyclic-P | H2O4P,O3P | 5'-OH | O-3P-1 | CpA 70-90%; CpG/CpU 80-100%; UpA 70-90%; UpU/ApU 14-30%; CpC 0% in the LC-MS study. Waters identifies Cp[A/U/G] and UpA as primary sites. | E034,E015 |
| RNase MC1 | Cleaves at the 5' side of U; ApU/CpU/UpU preferred, GpU nearly resistant. | \|U (full) | Mixed: linear 3'-P and 2',3'-cyclic-P | H2O4P,O3P | 5'-OH | O-3P-1 | ApU 70-100%; CpU 80-100%; UpU 45-60%; GpU 1.0-1.5%; CpA 10-20%. | E015 |
| colicin E5 | Sequence/structure-specific tRNase at anticodon positions 34-35; Q-U/G-U context. | G\|U (full) | 2',3'-cyclic-P | O3P | 5'-OH | O-3P-1 | Cleaves susceptible tRNAs at positions 34-35; not transcript-wide. | E016 |
| RNase CL3 | Chicken-liver cytidine-preferential ssRNA endoribonuclease. | Not modeled | Not established | | Not established | | C-containing to U-containing bond discrimination was 61:1 in limit digests of 5S/5.8S rRNA. | E024 |
| RNase L | 2-5A-activated, metal-independent ssRNA endoribonuclease; predominantly after U. | Not modeled | 2',3'-cyclic-P | O3P | 5'-OH | O-3P-1 | Preference approximately UU > UA >> UG > UC; magnitude depends strongly on local structure. | E019,E026 |
| RNase E | Low-specificity cleavage in AU-rich ssRNA; 5'-monophosphate-assisted pathway. | Not modeled | Not established | | Not established | | Prefers >=3 unpaired 5' nucleotides; optimal spacing about 8 nt; U/A/G-rich constructs diffused 2-3x faster than C-rich RNA in one assay. | E023,E027 |
| RNase G | RNase E homolog; low-specificity cleavage in AU-rich ssRNA. | Not modeled | Not established | | Not established | | Prefers >=3 unpaired 5' nucleotides; optimal spacing about 6 nt; more impeded than RNase E with <4 unpaired nucleotides. | E023 |

Biological product chemistry is not known for every enzyme. A blank formula in the implementation table means the reader uses the configured default; a blank biological-product field means the cited literature did not establish that terminus.

# Part II. mzLib implementation and resulting products

The following block is generated directly from the current clone's embedded rnases.tsv. It is retained as a text-format comparison against Part I.

```
Name | Motif | Specificity | ThreePrimeTerminusRemainder | FivePrimeTerminusRemainder | Notes
RNase T1 | G| | full | H2O4P | O-3P-1 | G, ssRNA
RNase T2 | G|,C|,A|,U| | full | H2O4P | O-3P-1 | ssRNA
RNase A | C|,U| | full | H2O4P | O-3P-1 | ssRNA
top-down |  | none
RNase 1 | G|,C|,A|,U| | full | H2O4P | O-3P-1 | Mammalian RNase 1, slight pyrimidine preference
RNase PhyM (>= 7M urea) | A|,U| | full | O3P |  | Up-N/Ap-N; almost exclusive and uniform under 7 M urea
RNase PhyM (< 7M urea) | A|,U|,G| | full | O3P |  | Up-N/Ap-N/Gp-N
RNase 4 | U|A,U|G | full | H2O4P,O3P | O-3P-1 | U-A slightly preferred over U-G; ssRNA; mixed 3' products
RNase U2 | G|,A| | full | H2O4P | O-3P-1 | Purines, A preferred over G
Cusativin | C[C]|,U|A,X|U | full | H2O4P,O3P | O-3P-1 | Primary sites Cp[A/U/G] and UpA; CpA 70-90%; CpG/CpU 80-100%; CpC resistant; mixed 3' products; Waters 720008488
RNase_MC1 | [G]|U | full | H2O4P,O3P | O-3P-1 | ApU 70-100%; CpU 80-100%; UpU 45-60%; GpU 1.0-1.5%; mixed 3' products
colicin_E5 | G|U | full | O3P | O-3P-1 | Sequence/structure-specific tRNase at anticodon positions 34-35
RNase I | G|,C|,A|,U| | full | H2O4P | O-3P-1 | E. coli RNase I; largely sequence-independent ssRNA endonuclease
non-specific | X| | full | H2O4P | O-3P-1 |
singleN |  | SingleN | H2O4P | O-3P-1 |
singleC |  | SingleC | H2O4P | O-3P-1 |
```

## Implementation checks

The | markers above render the tab boundaries for readability; the six-column order is unchanged.

| Check | Current behavior | Evidence |
|---|---|---|
| Formula defaults | Omitted formula fields become singleton lists: 3' H2O4P and 5' O-3P-1. | Rnase.cs lines 9-21; E001 |
| List-valued termini | ThreePrimeTerminusRemainder and FivePrimeTerminusRemainder are IList values. | Rnase.cs lines 12-21; E001 |
| TSV parsing | RnaseDictionary resolves both formula columns, splits comma-separated formulas, and constructs Rnase with both lists. | RnaseDictionary.cs lines 293-397; E003 |
| Multiple products | Internal products enumerate the Cartesian product of the 3' and 5' remainder lists. | Rnase.cs GetDigestedTermini; E001 |
| RNase 4 | H2O4P,O3P encodes two alternative 3' products; this represents alternatives, not probabilities. | rnases.tsv; E011,E018 |
| PhyM | Two condition-specific rows: A/U at >=7 M urea and A/U/G below 7 M urea. The blank 5' field therefore uses the default in code, although the biological 5' terminus is unreported. | rnases.tsv; E010,E029 |
| Backward compatibility | Existing three-column files remain valid because formula columns are optional and constructor defaults preserve prior behavior. | RnaseDictionary.cs; E001,E003 |

Worked mapping. For an internal cut in A-C|G-U, the default expected pair is upstream A-C with 3' remainder H2O4P and downstream G-U with 5' remainder O-3P-1. If the RNase supplies O3P as an alternative, the upstream product has both formula alternatives. Values on both sides produce all pairwise combinations.